

PRODUCT LITERATURE

MVT TABLET

Each tablet contains

Ferrous Sulphate	30.00 mg
Sodium Ascorbate	25.00 mg
Nicotinamide	20.00 mg
Thiamine Mononitrate	10.00 mg
Riboflavine	3.00 mg
Calcium Panthothenate	2.00 mg
Pyridoxine	2.00 mg
Vitamin E	2.00 mg
Folic Acid	1.00 mg
Vitamin A	5000 iu
Vitamin D	400 iu
Cyanocobalamin	2.00 µg

Description

Round brown-coloured sugar-coated tablet

Indication

Dietary supplement

Recommended Dose

Adult: 1 tablet daily

Route of administration

Oral.

Contraindication

Risk-benefit should be considered when the following medical problems exist:

- Thiamine
 - Sensitivity to thiamine
 - Wernicke's encephalopathy (intravenous glucose loading may precipitate or worsen this condition in thiamine-deficient patients; thiamine should be administered prior to glucose)
- Pyridoxine
 - Sensitivity to pyridoxine
- Cyanocobalamin
 - Sensitivity to cyanocobalamin
- Nicotinamide
 - Arterial bleeding or hemorrhage or Glaucoma (these conditions may be exacerbated)
 - Diabetes mellitus (large doses of niacin may cause impaired glucose tolerance; adjustment of diet and/or hypoglycemic therapy may be necessary)
 - Gout (large doses may cause hyperuricemia)
 - Hepatic disease (large doses may cause hepatic damage,
 - Hepatobiliary disease, history of or Jaundice, history of (patients with a past history of these medical problems should be closely observed during niacin extended-release therapy; frequent monitoring of liver function tests and blood glucose is recommended)
 - Hypotension (may worsen due to vasodilating effects of niacin)
 - Peptic ulcer (large doses may activate peptic ulcer)
 - Renal dysfunction (caution is advised when niacin extended-release is used in patients with renal dysfunction; increased drug exposure may result due to decreased renal elimination of niacin and metabolites)
- Ascorbic Acid
 - Diabetes mellitus (possible interference with glucose determinations by very high doses of ascorbic acid)
 - Glucose-6-phosphate dehydrogenase (G6PD) deficiency (high doses of ascorbic acid may cause hemolytic anemia)
 - Hemochromatosis or Sideroblastic anemia or Thalassemia (high doses of ascorbic acid may increase iron absorption)
 - Hyperoxaluria or oxalosis or Renal stones, history of (risk of hyperoxaluria and possible precipitation of oxalate stones in urinary tract after high doses of ascorbic acid)
 - Sensitivity to ascorbic acid or sodium ascorbate
- Folic Acid
 - Pernicious anemia (folic acid will correct hematologic abnormalities but neurologic problems will progress irreversibly; doses of folic acid greater than 0.4 mg per day are not recommended until pernicious anemia has been ruled out, except during pregnancy and lactation)
 - Sensitivity to folic acid
- Ferrous Sulphate
 - Alcoholism, active or in remission (alcohol may increase absorption and hepatic storage of iron and increase iron toxicity)
 - Allergies or Asthma (increased risk of hypersensitivity reactions with parenteral administration)
 - Hepatitis or hepatic function impairment or Kidney disease, acute, infectious (may cause an accumulation of iron)
 - Intestinal tract inflammatory conditions, such as enteritis, colitis, diverticulitis, and ulcerative colitis or Peptic ulcer (may be exacerbated with oral iron dosage forms)
 - Sensitivity to iron-caution is recommended also in patients receiving repeated blood transfusions because the addition of high erythrocytic iron content may produce iron overload.
- Vitamin D
 - Arteriosclerosis or Cardiac function impairment (conditions may be exacerbated due to possibility of hypercalcemia and elevated serum cholesterol concentrations)
 - Hyperphosphatemia (risk of metastatic calcification; dietary phosphate restriction or administration of intestinal phosphate binders is recommended to produce normal serum phosphorus concentrations)
 - Hypersensitivity to effects of vitamin D (may be involved in causing idiopathic hypercalcemia in infants)
 - Renal function impairment (toxicity may occur in patients receiving vitamin D for nonrenal problems, although toxicity is also possible during treatment of renal osteodystrophy because of increased requirements and decreased renal function)
 - Sarcoidosis, and possibly other granulomatous diseases (increased sensitivity to effects of vitamin D)
- Vitamin A
 - Alcoholism, chronic or Cirrhosis or Hepatic disease or Viral hepatitis (use of vitamin A in these conditions may potentiate hepatotoxicity; however, this may not apply in cases of chronic cholestatic liver disease with accompanying vitamin A malabsorption)
 - Renal failure, chronic (serum vitamin A concentrations are increased)
 - Sensitivity to vitamin A
- Vitamin E
 - Hypoprothrombinemia due to vitamin K deficiency (may be aggravated by doses of vitamin E greater than 400 U a day)
 - Sensitivity to vitamin E

Except under special circumstances, this medication should not be used when the following medical problems exist:

- Ferrous Sulphate
 - Hemochromatosis or Hemosiderosis (existing iron overload may be increased)
 - Other anemic conditions, unless accompanied by iron deficiency (some conditions, such as hemolytic anemia or thalassemia, may cause excess storage of iron)
 - Porphyria cutanea tarda (may be caused by hepatic accumulation of iron, as in iron overload)
 - Note: These medical problems are contraindications for niacin extended-release due to the higher doses used to treat hyperlipidemia. These medical problems are relative contraindications (risk-benefit) for niacin used for prophylaxis and treatment of niacin deficiency.
 - Arterial bleeding (niacin extended-release is contraindicated in patients with arterial bleeding)

- Hepatic dysfunction, active or Transaminases, serum, unexplained elevations (niacin extended-release is contraindicated in patients with significant or unexplained hepatic dysfunction)
 - Hypersensitivity to niacin or niacinamide
 - Peptic ulcer disease, active (niacin extended-release is contraindicated in patients with active peptic ulcer disease)
- Riboflavin / Calcium Panthothenate
 - Data not available
- Vitamin D
 - Hypercalcemia
 - Hypervitaminosis D
 - Renal osteodystrophy with hyperphosphatemia (risk of metastatic calcification; however, vitamin D therapy can begin once serum phosphate levels have stabilized)
- Vitamin A
 - Hypervitaminosis A

Warning and Precautions

Pregnancy/Reproduction

- Thiamine / Pyridoxine / Riboflavine / Nicotinamide / Calcium Panthothenate / Ascorbic Acid / Folic Acid / Vitamin A / Vitamin D / Vitamin E
 - Problems in humans have not been documented with intake of normal daily recommended amounts.
- Pyridoxine
 - However, exposure to large doses of pyridoxine in utero may result in a pyridoxine dependency syndrome in the neonate.
 - Adequate and well-controlled studies in humans and animals have not been done at doses typically used for lipid disorders.
- Nicotinamide
 - If a women receiving niacin extended-release for primary hypercholesterolemia (Types IIa or IIb) becomes pregnant, the drug should be discontinued. If a woman being treated with niacin extended-release for hypertriglyceridemia (Types IV or V) conceives, the benefits and risks of continued therapy should be assessed per individual.
- Cyanocobalamin
 - Adequate and well-controlled studies have not been done in humans. Diagnostic testing should be postponed, if possible, until after delivery since cyanocobalamin crosses the placenta and is taken up by the fetus. To avoid the possibility of radiation exposure to the fetus, in those circumstances w here the patient's pregnancy status is uncertain, a pregnancy test will help to prevent inadvertent administration of this preparation during pregnancy. Studies have not been done in animals. FDA Pregnancy Category C.
- Ascorbic Acid
 - Ascorbic acid crosses the placenta. Ingestion of large quantities of ascorbic acid daily throughout pregnancy may possibly harm the fetus. Studies have not been done in animals.
- Folic Acid
 - Folic acid crosses the placenta. However, adequate and well-controlled studies in humans have not shown that folic acid causes adverse effects on the fetus. Some studies have found that folic acid supplementation alone or in combination with other vitamins given before conception and during early pregnancy may reduce the incidence of neural tube defects in infants. FDA Pregnancy Category A.
- Ferrous Sulphate
 - In the first trimester of pregnancy, adequate iron intake is usually obtained from a proper diet; however, in the second and third trimesters, when iron deficiency is more prevalent because of greatly increased requirements, iron supplements may be recommended. However, some clinicians prefer to evaluate the patient before giving routine iron supplementation. Studies in humans have not been done, and problems in humans have not been documented with intake of normal daily recommended amounts. Studies in animals have not been done.
- Vitamin A
 - Vitamin A crosses the placenta to only a limited extent. Fetal abnormalities (including urinary tract malformations), growth retardation, and early epiphyseal closure have been reported in children whose mothers took excessive amounts during pregnancy. Daily amounts from supplements exceeding 1800 RE (6000 Units) are not recommended because of potential fetotoxicity. Maternal overdosage in animals has been reported to result in central nervous system (CNS) malformations, as well as malformations of the spinal column, rib cage, heart, eye, palate, and genitourinary tract of the fetus.
- Vitamin D
 - Fertility; Paricalcitol had no effect on the fertility of rats at intravenous doses up to 20 micrograms/kg (mcg/kg) per dose (equivalent to 13 times the highest recommended human dose (0.24 mcg/kg) based on body surface area (mg/m²)) Doxercalciferol had no effect on the fertility of rats at doses up to 2.5 micrograms/kg/day (mcg/kg/day) (approximately 3 times and less than the maximum recommended human dose of 60 mcg/week based on mg/m² body surface area)
 - Pregnancy; Problems in humans have not been documented with intake of normal daily recommended amounts. There are insufficient data on acute and chronic vitamin D toxicity in pregnant women. Maternal hypercalcemia during pregnancy in humans may be associated with increased sensitivity to effects of vitamin D, suppression of parathyroid function, or a syndrome of peculiar (elfin) facies, mental retardation, and congenital aortic stenosis in infants. Overdosage of vitamin D has been associated with fetal abnormalities in animals.

Interactions with Other Medicaments

- Thiamine / Cyanocobalamin / Calcium Panthothenate
 - Data not available
- Pyridoxine:
 - Cycloserine or Ethionamide or Hydralazine or Immunosuppressant, such as: Azathioprine, Chlorambucil, Corticosteroids, Corticotropin (ACTH), Cyclophosphamide, Cyclosporine, Mercaptopurine, or Isoniazid or Penicillamine (may cause anemia or peripheral neuritis by acting as pyridoxine antagonists or increasing renal excretion of pyridoxine; recommended intakes for pyridoxine may be increased in patients receiving these medications)
 - Estrogens or Contraceptives, estrogen-containing, oral (may increase recommended intakes for pyridoxine)
 - Levodopa (concurrent use with pyridoxine is not recommended since levodopa's antiparkinsonian effects are reversed by as little as 5 mg of pyridoxine orally; this problem does not occur with the carbidopa-levodopa combination)
- Riboflavine
 - Alcohol (impairs intestinal absorption of riboflavin)
 - Antidepressants, tricyclic or Phenothiazines (requirements for riboflavin may be increased in patients receiving these medications)
 - Probenecid (concurrent use decreases gastrointestinal absorption of riboflavin; requirements for riboflavin may be increased in patients receiving probenecid)
- Nicotinamide
 - Alcohol (niacin extended-release and concomitant alcohol may result in increased adverse effects, such as flushing and pruritus)
 - Anticoagulants (caution should be used with concomitant use with niacin extended-release; prothrombin time and platelet count monitoring is recommended)
 - Antihypertensives (niacin may potentiate the effects of ganglionic blocking agents and vasoactive therapy resulting in postural hypotension)

- Aspirin (concomitant use may decrease the metabolic clearance of nicotinic acid)
 - Bile Acid Sequestrants (as great an interval as possible should elapse between the ingestion of bile acid-binding resins and the administration of niacin)
 - Chenodiol or Ursodiol (effect may be decreased when chenodiol or ursodiol is used concurrently with antihyperlipidemics, which tend to increase cholesterol saturation of bile)
 - HMG-CoA reductase inhibitors (concurrent use with niacin may be associated with an increased risk of rhabdomyolysis and acute renal failure; combined therapy with lovastatin, pravastatin, or simvastatin should include careful monitoring for symptoms of myopathy or rhabdomyolysis)
 - Vasoactive drugs, such as: Adrenergic blocking agents or Calcium channel blockers, or Nitrates (caution should be used when niacin extended-release is used in patients with unstable angina or in the acute phase of myocardial infarction, especially patients also receiving vasoactive drugs; may result in increased risk of hypotension)
 - Vitamins or Other supplements containing niacin or related compounds such as nicotinamide (may potentiate the adverse effects of niacin extended-release)
- Ascorbic Acid
 - Large doses of ascorbic acid (but not sodium ascorbate) may lower urinary pH and cause renal tubular reabsorption of acidic medications with concurrent administration; alkaline medications may exhibit decreased reabsorption.
 - Anticoagulants, coumarin- or indandione-derivative (doses of 10 grams or more a day of ascorbic acid have been reported to impair gastrointestinal absorption of the anticoagulant)
 - Cellulose sodium phosphate (concurrent use may result in metabolism of ascorbic acid to oxalate)
 - Deferoxamine (concurrent use with ascorbic acid may enhance tissue iron toxicity, especially in the heart, causing cardiac decompensation; therefore, this regimen should be used with caution in older patients; the need for ascorbic acid supplementation should be completely documented by measurements of iron excretion before and after supplements, and the oral dose of ascorbic acid should be given an hour or two after the deferoxamine infusion has been initiated when adequate concentrations of deferoxamine have been achieved)
 - Disulfiram (concurrent use with ascorbic acid, especially with chronic use or high doses, may interfere with the disulfiram-alcohol interaction)
- Folic Acid
 - Analgesics, long-term use or Anticonvulsants, hydantoin or Carbamazepine or Estrogens or Oral contraceptives or Phenobarbital or Primidone (requirements for folic acid may be increased in patients receiving these medications)<

Pregnancy and Lactation

Refer to Warning and Precautions

Side Effects

Those indicating need for medical attention

- Incidence rare:
 - Thiamine
Anaphylactic reaction (coughing; difficulty in swallowing; hives; itching of skin; swelling of face, lips, or eyelids; wheezing or difficulty in breathing); usually after a large intravenous dose
 - Pyridoxine
Doses of 200 mg per day for over 30 days have been reported to produce a pyridoxine dependency syndrome. High doses of pyridoxine (2 to 6 grams per day) taken for several months have caused a severe sensory neuropathy, progressing from unstable gait and numb feet to numbness and clumsiness of hands. This condition seems to be reversible on withdrawal of pyridoxine, although some residual weakness has been observed.
 - Riboflavine
While toxicity with high doses of riboflavin has not been reported, high doses of other water-soluble vitamins have been known to cause problems. Large doses of riboflavin may cause yellow discoloration of urine.
 - Nicotinamide
Allergic reaction, anaphylactic (skin rash or itching; wheezing); after intravenous administration
 - Folic Acid
Allergic reaction, specifically bronchospasm (shortness of breath; troubled breathing; tightness of chest; wheezing), erythema (reddened skin), fever, general malaise (general weakness or discomfort), skin rash or itching
Note: Side effects including abdominal bloating and gas, anorexia, confusion, depression, difficulty in concentrating, excitement, irritability, impaired judgment, nausea, trouble in sleeping, and unpleasant taste were reported in a study of patients taking 15 mg daily.
- Incidence dose-related:
 - Ascorbic Acid:
Kidney stones, oxalate (side or lower back pain)
Note: Occasionally, prolonged doses of ascorbic acid in excess of 1 g per day have been reported to cause an increase in urinary oxalate, which may cause precipitation of oxalate stones in the urinary tract in patients with renal disease, especially those on hemodialysis, or in patients with a history of renal stones. However, studies have not found an increase in urinary oxalate formation with high doses of ascorbic acid.
- Incidence more frequent:
 - Ferrous Fumarate
Abdominal or stomach pain, cramping, or soreness

Those indicating need for medical attention only if they continue or are bothersome

- Incidence less frequent with niacin only:
 - Feeling of warmth flushing or redness of skin, especially on face and neck headache pain, Abdominal, rash, rhinitis (stuffy nose; runny nose; sneezing)
- With high oral doses with niacin only:
 - Cardiac arrhythmias (unusually fast, slow, or irregular heartbeat), diarrhea, dizziness or faintness, dryness of skin or eyes, hyperglycemia (frequent urination or unusual thirst) may occasionally be fatal, hyperuricemia (joint pain; side, lower back, or stomach pain; swelling of feet or lower legs), myalgia (fever; muscle aching or cramping; unusual tiredness or weakness), nausea or vomiting, peptic ulcer, aggravation of (stomach pain), Pruritus (itching of skin); may be severe.
 - Note: Rarely, along with markedly elevated creatine kinase (CK) concentrations, fever, muscle aching or cramping, or unusual tiredness or weakness may be symptoms of myositis or rhabdomyolysis; incidence may be increased in patients treated concurrently with lovastatin, pravastatin, or simvastatin.

Symptoms and Treatment of Overdose

- Thiamine / Pyridoxine / Riboflavine / Nicotinamide / Cyanocobalamin / Calcium Panthothenate / Ascorbic Acid / Folic Acid
 - Data not available

- Clinical effects of overdose**

The following effects have been selected on the basis of their potential clinical significance (possible signs and symptoms in parentheses w here appropriate); not necessarily inclusive:

Acute effects

- Early symptoms of acute iron toxicity
- Diarrhea, sometimes containing blood,*
- fever,*
- nausea, severe,*
- stomach pain or cramping, sharp,*
- vomiting, severe, sometimes containing blood*
- Note:** Early symptoms may not be evident for up to 60 minutes or longer; if overdose is suspected, emergency room treatment should not be delayed for evidence of symptoms, but should begin immediately. Early signs may also include increased blood glucose and leukocytosis.

A latency period lasting from 2 to about 48 hours after ingestion may occur between the 2 symptomatic phases.

During this time, the patient may appear to improve clinically

Late symptoms of acute iron toxicity

- Bluish-colored lips, fingernails, palms of hands*
- Drowsiness*
- pale, clammy skin*
- seizures*
- shallow and rapid breathing-* due to acidosis
- unusual tiredness or weakness*
- weak and fast heartbeat*

Note: Late signs may also include metabolic acidosis, hypotension, hypoglycemia, hepatic injury or failure, cardiovascular collapse, and gastrointestinal scarring.

- For treatment of acute overdose:**
Overdose of iron by any route may be fatal, and immediate treatment is essential. **Overdoses of ingested iron can be fatal, especially in small children.** Serious poisoning may result in small children from ingestion of 3 or 4 ferrous sulfate tablets (200 mg of elemental iron). Acute overdose of iron requires immediate medical treatment that should be completed as soon as possible following ingestion. After one hour, excessive systemic absorption of iron and possible erosion of stomach and intestinal tissues complicate evacuative and supportive procedures. Transport of patient to emergency room should not be delayed. If syrup of ipecac has been administered, emesis may require up to 30 minutes or even a repeat dose. However, patient transport must not wait for emetic effect. Overdose symptoms may be delayed (10 to 60 minutes or longer) because of many intervening factors such as the iron salt taken, amount of food in stomach, and size of dose. To decrease absorption (after oral ingestion):
Inducing emesis with syrup of ipecac or lavaging with sodium bicarbonate if patient is comatose or having convulsions may be used, depending on the patient's condition. If intact radiopaque tablets are visualized on x-ray, repeating lavage may be necessary.

- Monitoring:
Laboratory studies on heparinized blood should include serum iron, hemoglobin, hematocrit, electrolytes, blood gases and blood glucose, total iron-binding capacity (TIBC), complete blood count, blood type, and cross-match.
Serum iron determinations should be repeated. Serum drawn early (within 2 hours of ingestion) may have artificially high concentrations of iron. Peak serum concentrations are reached about 6 hours after ingestion. However, achievement of peak serum concentrations is delayed if the iron was in extended-release form, if the patient had a significant amount of food in his/her stomach, and if the dose of iron was large.

- Specific treatment:
 - Fluid and electrolyte balance must be maintained. Acidosis may be corrected with intravenous sodium bicarbonate.
 - Antidote; Deferoxamine, administered slowly, intravenously or intramuscularly, is used in more severe iron toxicity, when symptoms are other than minimal vomiting or diarrhea. Deferoxamine chelates iron to form a red soluble ferric complex (ferrioxamine) that is excreted in the urine. Children with a history of ingesting > 40 mg of elemental iron per kg of body weight, or if serums iron determinations and TIBC are not available, should receive an intramuscular test dose of deferoxamine, regardless of symptoms. If the urine turns an orange-rose (vin ros;) color, deferoxamine should be continued intravenously. However, a negative test dose does not rule out iron toxicity, since false negative tests have been reported with deferoxamine. When results of serum iron determinations and TIBC are available, dosing should continue, if necessary.
 - Avoid deferoxamine in patients who have developed renal failure.
 - Dialysis is of no value in removing serum iron alone, but may be used to increase excretion of the iron-deferoxamine complex, and is indicated in the presence of anuria

Packing

Packed in plastic container of 30 tablets and 100 tablets.

Storage Conditions

Store below 30°C in a dry place, protected from direct light.

Keep away from reach of children.

Expiry Date

2 years from date of manufacture

Product Registration Holder / Distributor:

ZONTRON PHARMACEUTICALS SDN.BHD. (445695-T)

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Malaysian Drug Registration No. :

MAL19950114XC

Date of Revision:

21/05/2020